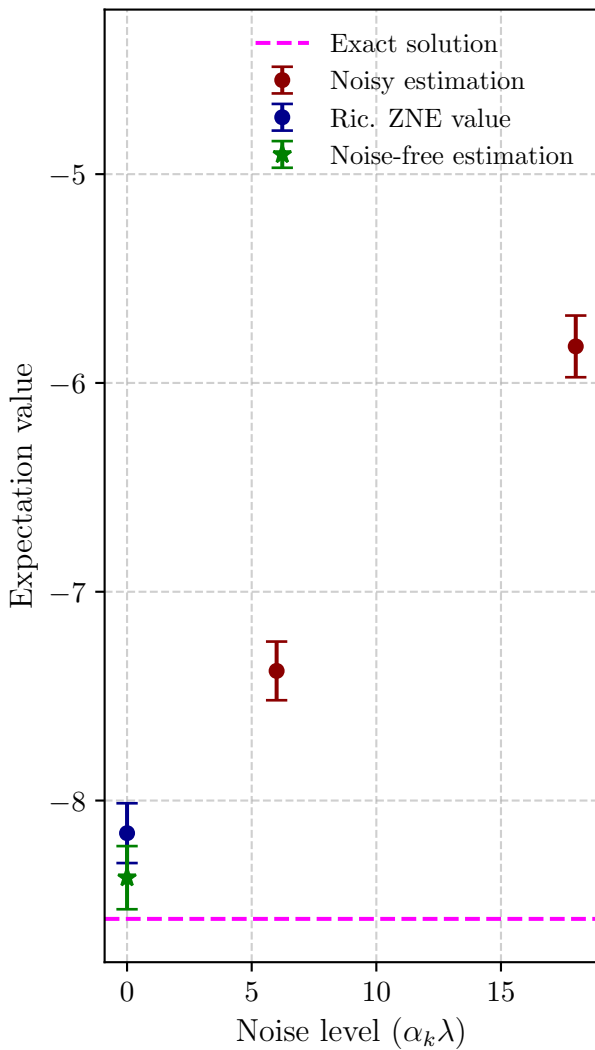
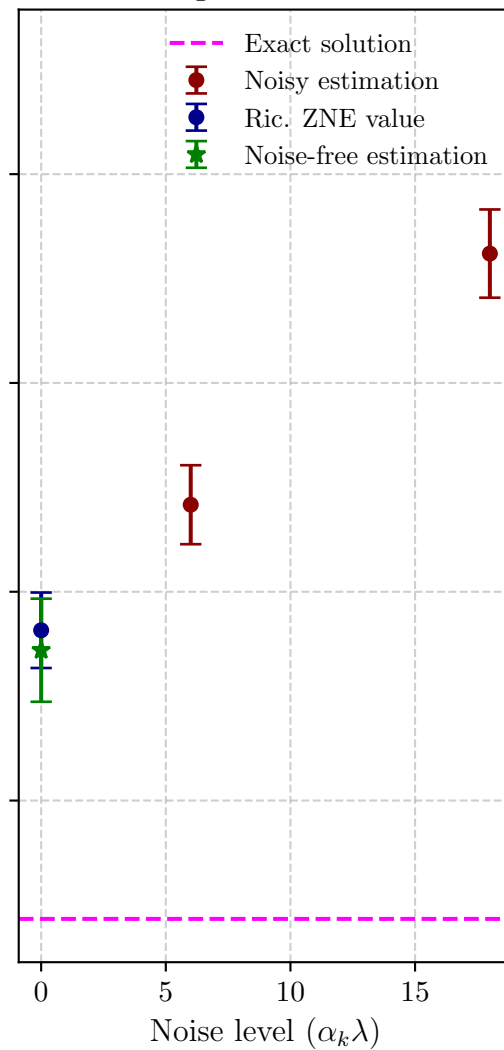


| Description                     | Simulation           | Mean $\pm$ Std         |
|---------------------------------|----------------------|------------------------|
| xy-ric2                         | vqe                  | -7.37909 $\pm$ 0.14052 |
| xy-ric2                         | redundant (noise=6)  | -7.37909 $\pm$ 0.14052 |
| xy-ric2                         | redundant (noise=18) | -5.82492 $\pm$ 0.14775 |
| xy-ric2                         | zne                  | -8.15618 $\pm$ 0.14317 |
| xy-ric2                         | noiseoff             | -8.36934 $\pm$ 0.15125 |
|                                 |                      |                        |
| ising-ric2                      | vqe                  | -6.58315 $\pm$ 0.18915 |
| ising-ric2                      | redundant (noise=6)  | -6.58315 $\pm$ 0.18915 |
| ising-ric2                      | redundant (noise=18) | -5.38044 $\pm$ 0.21133 |
| ising-ric2                      | zne                  | -7.18451 $\pm$ 0.18074 |
| ising-ric2                      | noiseoff             | -7.28002 $\pm$ 0.24691 |
|                                 |                      |                        |
| heisenberg-ric2                 | vqe                  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric2                 | redundant (noise=6)  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric2                 | redundant (noise=18) | -4.59153 $\pm$ 0.17228 |
| heisenberg-ric2                 | zne                  | -6.78892 $\pm$ 0.19421 |
| heisenberg-ric2                 | noiseoff             | -6.92055 $\pm$ 0.19767 |
|                                 |                      |                        |
| xy-ric3-uniform-scaling         | vqe                  | -7.37909 $\pm$ 0.14052 |
| xy-ric3-uniform-scaling         | redundant (noise=6)  | -7.37909 $\pm$ 0.14052 |
| xy-ric3-uniform-scaling         | redundant (noise=18) | -5.82492 $\pm$ 0.14775 |
| xy-ric3-uniform-scaling         | redundant (noise=30) | -4.62559 $\pm$ 0.15660 |
| xy-ric3-uniform-scaling         | zne                  | -8.28925 $\pm$ 0.14534 |
| xy-ric3-uniform-scaling         | noiseoff             | -8.36934 $\pm$ 0.15125 |
|                                 |                      |                        |
| ising-ric3-uniform-scaling      | vqe                  | -6.58315 $\pm$ 0.18915 |
| ising-ric3-uniform-scaling      | redundant (noise=6)  | -6.58315 $\pm$ 0.18915 |
| ising-ric3-uniform-scaling      | redundant (noise=18) | -5.38044 $\pm$ 0.21133 |
| ising-ric3-uniform-scaling      | redundant (noise=30) | -4.39924 $\pm$ 0.22365 |
| ising-ric3-uniform-scaling      | zne                  | -7.26757 $\pm$ 0.17710 |
| ising-ric3-uniform-scaling      | noiseoff             | -7.28002 $\pm$ 0.24691 |
|                                 |                      |                        |
| heisenberg-ric3-uniform-scaling | vqe                  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric3-uniform-scaling | redundant (noise=6)  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric3-uniform-scaling | redundant (noise=18) | -4.59153 $\pm$ 0.17228 |
| heisenberg-ric3-uniform-scaling | redundant (noise=30) | -3.46107 $\pm$ 0.17562 |
| heisenberg-ric3-uniform-scaling | zne                  | -6.91435 $\pm$ 0.19705 |
| heisenberg-ric3-uniform-scaling | noiseoff             | -6.92055 $\pm$ 0.19767 |
|                                 |                      |                        |
| xy-ric4-uniform-scaling         | vqe                  | -7.37909 $\pm$ 0.14052 |
| xy-ric4-uniform-scaling         | redundant (noise=6)  | -7.37909 $\pm$ 0.14052 |
| xy-ric4-uniform-scaling         | redundant (noise=18) | -5.82492 $\pm$ 0.14775 |
| xy-ric4-uniform-scaling         | redundant (noise=30) | -4.62559 $\pm$ 0.15660 |
| xy-ric4-uniform-scaling         | redundant (noise=42) | -3.69251 $\pm$ 0.16044 |
| xy-ric4-uniform-scaling         | zne                  | -8.31693 $\pm$ 0.14624 |
| xy-ric4-uniform-scaling         | noiseoff             | -8.36934 $\pm$ 0.15125 |
|                                 |                      |                        |
| ising-ric4-uniform-scaling      | vqe                  | -6.58315 $\pm$ 0.18915 |
| ising-ric4-uniform-scaling      | redundant (noise=6)  | -6.58315 $\pm$ 0.18915 |
| ising-ric4-uniform-scaling      | redundant (noise=18) | -5.38044 $\pm$ 0.21133 |
| ising-ric4-uniform-scaling      | redundant (noise=30) | -4.39924 $\pm$ 0.22365 |
| ising-ric4-uniform-scaling      | redundant (noise=42) | -3.59276 $\pm$ 0.22745 |
| ising-ric4-uniform-scaling      | zne                  | -7.28218 $\pm$ 0.17606 |
| ising-ric4-uniform-scaling      | noiseoff             | -7.28002 $\pm$ 0.24691 |
|                                 |                      |                        |
| heisenberg-ric4-uniform-scaling | vqe                  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric4-uniform-scaling | redundant (noise=6)  | -6.05646 $\pm$ 0.18070 |
| heisenberg-ric4-uniform-scaling | redundant (noise=18) | -4.59153 $\pm$ 0.17228 |
| heisenberg-ric4-uniform-scaling | redundant (noise=30) | -3.46107 $\pm$ 0.17562 |
| heisenberg-ric4-uniform-scaling | redundant (noise=42) | -2.58067 $\pm$ 0.17819 |
| heisenberg-ric4-uniform-scaling | zne                  | -6.94072 $\pm$ 0.19769 |
| heisenberg-ric4-uniform-scaling | noiseoff             | -6.92055 $\pm$ 0.19767 |
|                                 |                      |                        |

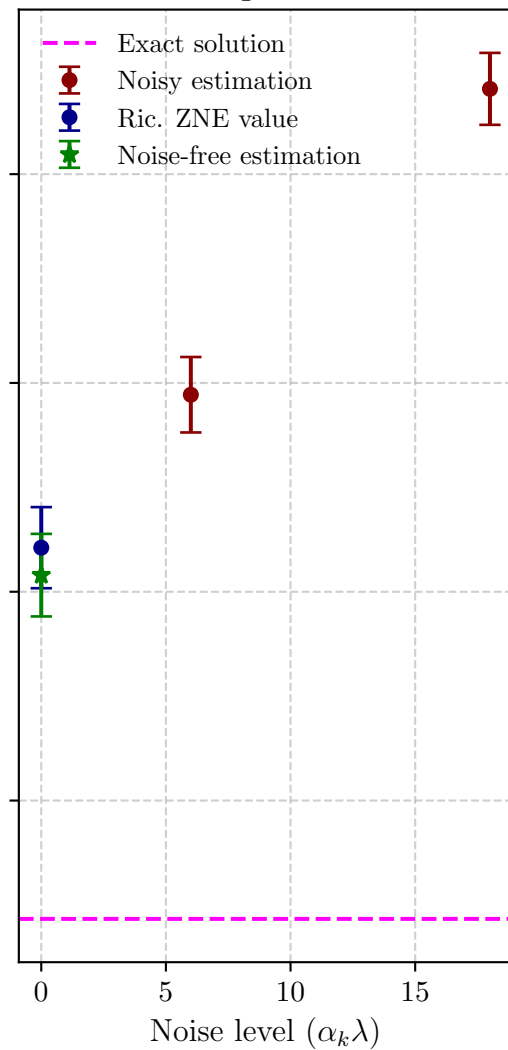
XY ansatz ric-2



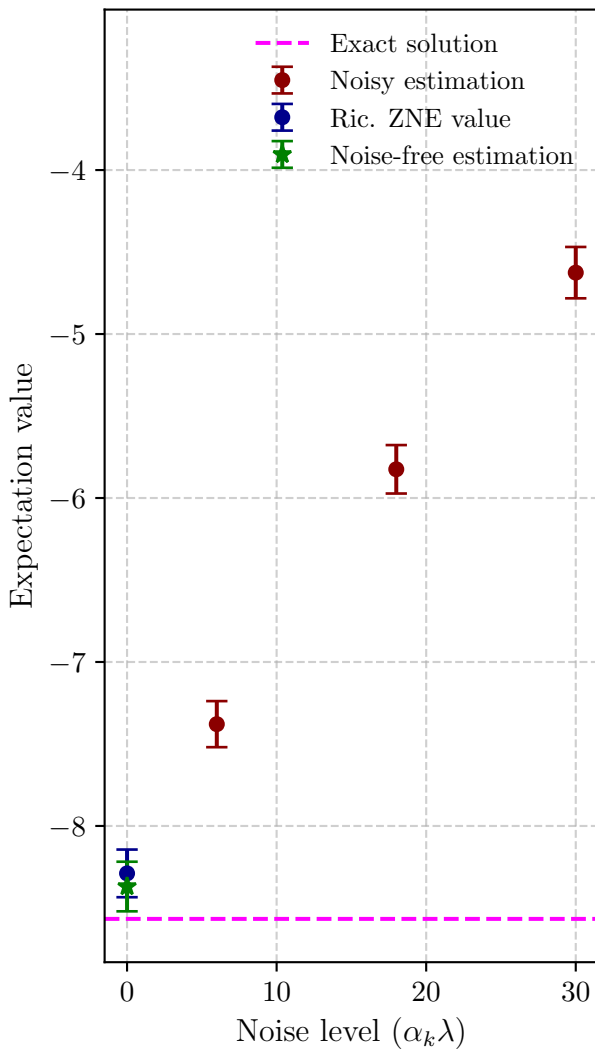
Ising ansatz ric-2



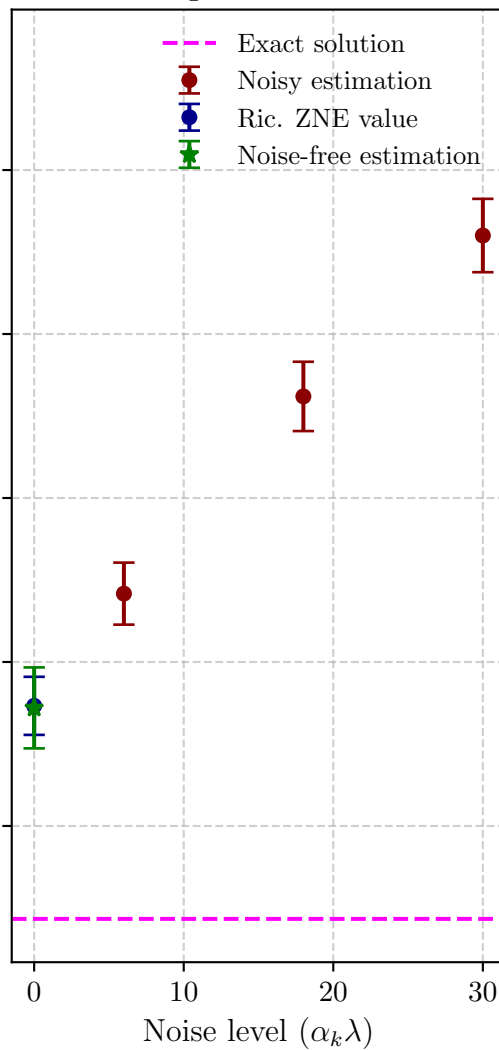
Heisenberg ansatz ric-2



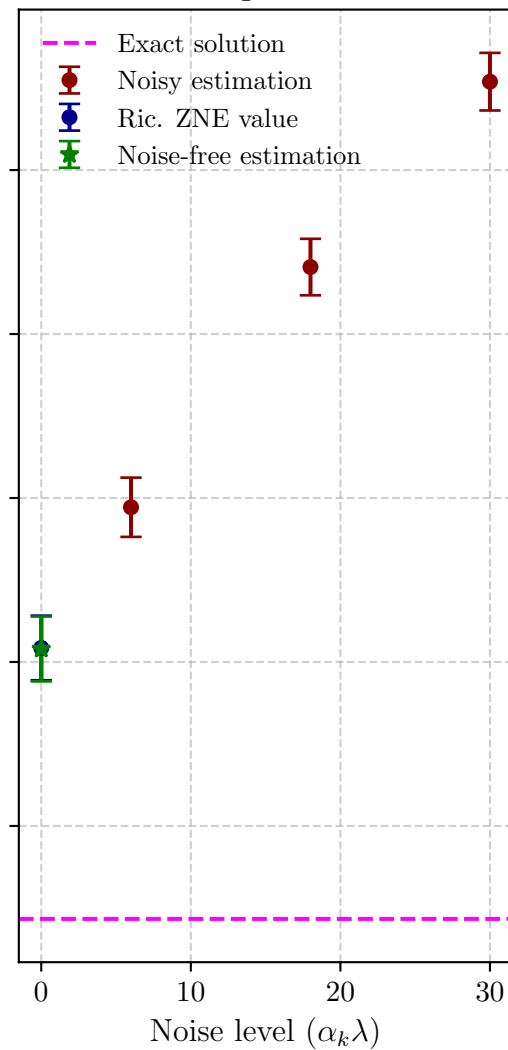
XY ansatz ric-3



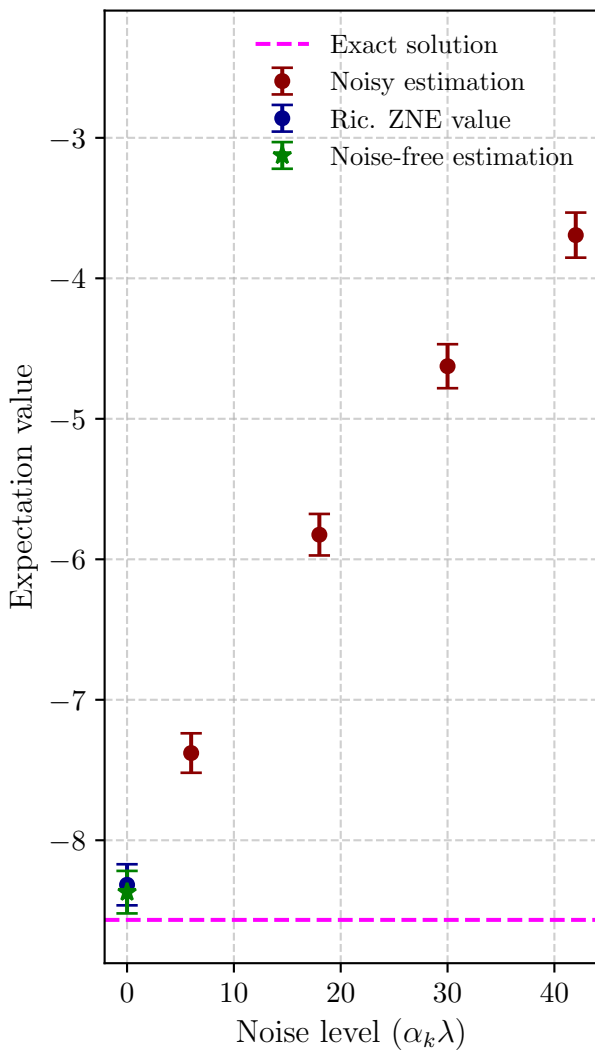
Ising ansatz ric-3



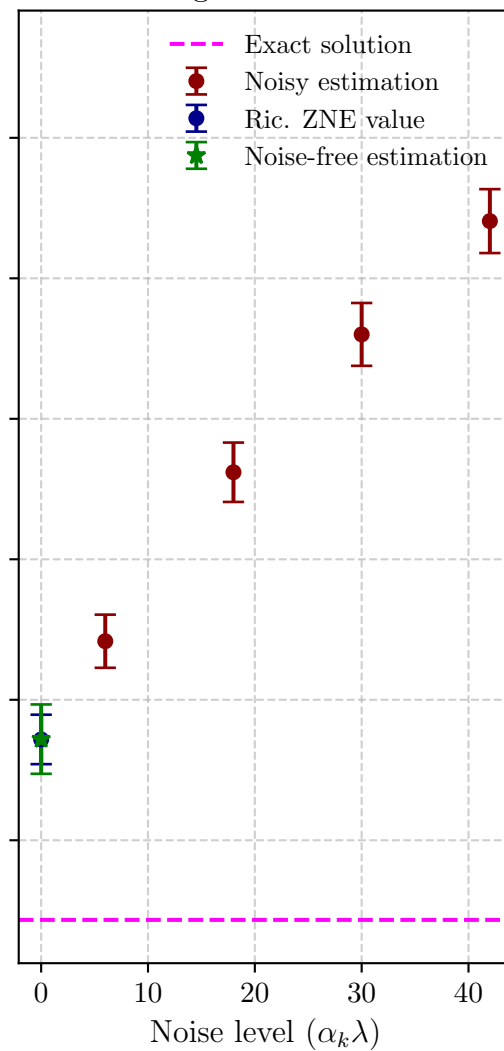
Heisenberg ansatz ric-3



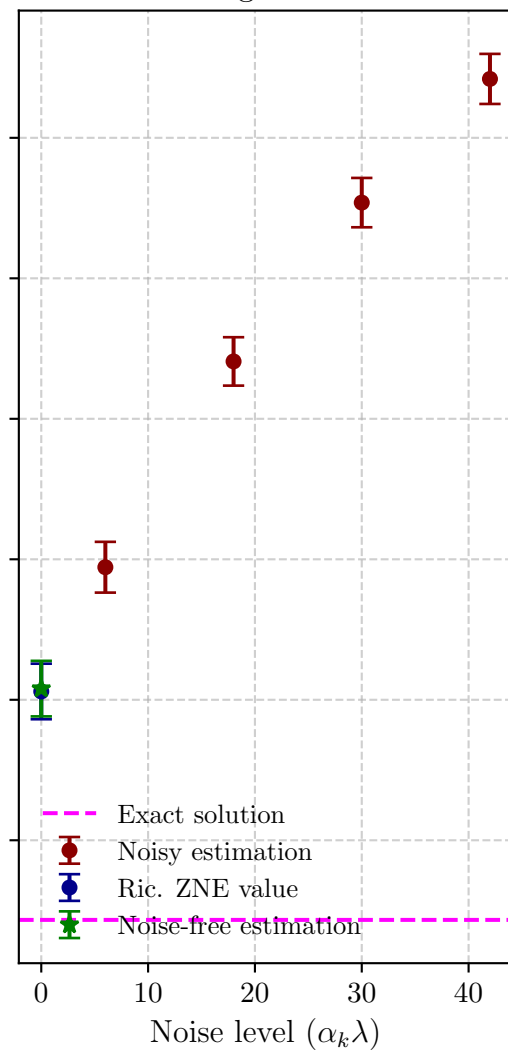
XY ansatz ric-4



Ising ansatz ric-4



Heisenberg ansatz ric-4



# XY VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | xy_noisefree_time_evo               |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | xy-iss                              |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | depolarizing                        |
| noise_profile.noise_prob                 | [0.001, 0.001, 0, 0]                |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 0, 0]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |

# ISING VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | ising_noisefree.time_evo            |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | ising                               |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | depolarizing                        |
| noise_profile.noise_prob                 | [0.001, 0.001, 0, 0]                |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 0, 0]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |

# HEISENBERG VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name.prefix                  | heisenberg_noisefree_time_evo       |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | heisenberg                          |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | depolarizing                        |
| noise_profile.noise_prob                 | [0.001, 0.001, 0, 0]                |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 0, 0]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |